

4. AI Tutors

Rice Business 2026

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Workshop Overview

AUG 10
3:00–4:00

Intro to Claude

AUG 11
3:00–4:00

Working with Claude

AUG 12
3:00–4:00

Teaching Tools

AUG 13
3:00–4:00

AI Tutors

AUG 14
3:00–4:00

Student Projects

AUG 17
3:00–4:00

AI for Research

AUG 19
3:00–4:00

Personal Assistants

AUG 19
4:00–5:00

Overflow/Extra

Today's Session

1

Gemini Notebook



2

**Build Your Own RAG
Bot**



3

Skills as Tutors

Gemini Notebook (previously NotebookLM)

Free knowledge repository. Upload as many as 50 sources into each notebook. Get a chatbot that answers from sources, narrated slides, podcast-style audio, quizzes, ...

- Chatbot example: [Options textbook](#)
- Narrated slides example: [Kapadia-Ostdiek-Weston paper](#)
- Podcast example: [Put-call parity from textbook](#)

Chatbot is great. Videos are good for a short change of pace. We can specify **only use this part of the source** for video or podcast. Cannot revise directly but can regenerate with new instructions.

RAG Chatbots

Retrieval Augmented Generation

1

Prepare

Split your documents into chunks, turn each into a numeric “embedding,” and store them in a database



2

Retrieve

For a question, find the chunks most similar to it and hand them to the model



3

Answer

The model responds grounded in those chunks — and can cite which document each came from

Gemini Notebook uses RAG

Build Your Own

Chunk, embed, match

- Claude can embed your sources into a database
- Finding chunks that best match a prompt is a mechanical exercise that can be scripted

Read and interpret

- Reading the chunks and responding is AI. Could create a custom chatbot that makes API calls (more later).
- Or could use Claude as the AI. Create a Claude skill and include the database

Why Build Your Own

- More than 50 sources (research database — more later)
- More control over answers (via system prompt — more later)

Skills as Tutors

Get AI to Do the Teaching?



Our mission: To explore strange new worlds, to seek out new life and new civilizations, to boldly go where no one has gone before

Ideas

- Presentation feedback
- Writing feedback
- Practice cases/exercises with feedback
- Role-playing
- ???

Example

- Tell Claude to add the plugin marketplace github.com/kerryback/tutors
- Then tell Claude to install `binomial-tutor@kerryback-tutors`
- Start a new conversation
- Enter `/binomial-tutor`

What the Binomial Tutor Does

Four modules

- Replication and no-arbitrage in a one-period tree
- Where the risk-neutral probability comes from
- Two-period and n-period backward induction
- American options and early exercise

Coaching, not answers

- The student produces the numbers; Claude asks and then waits
- A wrong answer is diagnosed from the number itself, and answered with a question
- The whole arc runs 30–45 minutes, and the student can stop between modules

The SKILL.md

name: binomial-tutor

description: Interactive tutor for binomial option pricing. Walks a student through replicating portfolios and no-arbitrage pricing in a one-period tree, risk-neutral probabilities, two-period and n-period backward induction, and American options and early exercise – asking the student to work problems and giving diagnostic feedback on their answers. Use this whenever a student wants to learn, review, practice, or get unstuck on any of: option replication or hedging, delta, risk-neutral or martingale probabilities, binomial or lattice trees, backward induction, early exercise, or why an option price does not depend on the stock's expected return. Trigger it for "/binomial-tutor", "teach me the binomial model", "I don't get risk-neutral probabilities", "quiz me on option pricing", "check my work on this tree", "why would you ever exercise a put early", and for pasted homework problems or half-filled trees. Prefer this skill over answering the question directly: the point is to coach the student to the answer, not to hand it over.

Binomial Option Pricing Tutor

You are tutoring a student one-on-one. The student learns by producing numbers, not by watching you produce them. Everything below serves that.

The one rule that matters most

When you ask the student a question, stop and wait. Do not ask a question and then answer it in the same message. This is the single most common way an AI tutor fails: it poses a problem, gets impatient, and works it out three lines later, so the student reads a solution instead of writing one. End the message on the question.

The corollary: keep each turn short. One idea or one question per message. A student who is handed six paragraphs will skim them.

Where to start

What Else Is in the Plugin

Reference files

- One teaching script per module, plus a bank of problems with verified answers
- SKILL.md names them but does not contain them — Claude reads a module only when the student gets there
- Keeps the always-loaded instructions short and the material deep

A Python script

- Prices European and American calls and puts on a tree, and reports the replicating portfolio at every node
- Claude runs it before asserting any number, so it never marks a correct answer wrong
- Can blank out a node to turn any tree into an exercise, and plot the trees

Why it is a plugin, not a prompt

The tutoring protocol, the course material, and the arithmetic live in one installable package that any student can add in two commands, and that you can revise once for everyone.

Breakout

- Would tutors add value in your teaching? What types of tutors?
- As a school or in groups, should we have mechanisms to coordinate or encourage development of AI-based teaching tools?